

CS S Review Guide

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Note: Review Guide are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

String Operation

Indexing & Slicing

```
s = "hello"
```

```
print(s[1])    # e
```

```
print(s[-1])   # o
```

```
print(s[1:4])  # ell
```

```
print(s[:2])   # he
```

```
print(s[2:])   # llo
```

```
print(s[:])    # hello
```

s[1]: the second character

s[-1]: last character

s[1:4]: includes index 1 up to but not including 4

s[:2]: start to index 2

s[2:]: from index 2 to end

s[:]: copy of the whole string

Common String Methods

.upper() → ALL CAPS

.lower() → all lowercase

.title() → First Letter Capitalized Each Word

find(), count(), replace()

```
s = "banana"
```

```
print(s.find("na"))    # 2
```

```
print(s.count("a"))    # 3
```

```
print(s.replace("na", "NA")) # baNANA
```

isupper() / islower()

```
s = "Hello"
```

```
print(s.isupper())     # False
```

```
print(s.islower())     # False
```

ASCII functions: ord() and chr()

```
print(ord('A'))    # 65
```

```
print(chr(97))     # a
```

```
print(chr(48))     # 0
```

The Internet, The Web and Electronic Commerce

Internet: massive network of networks, it's actually a physical network that's very interconnected

TCP/IP – Transmission Control Protocol (TCP) (Packing and Reassembling)

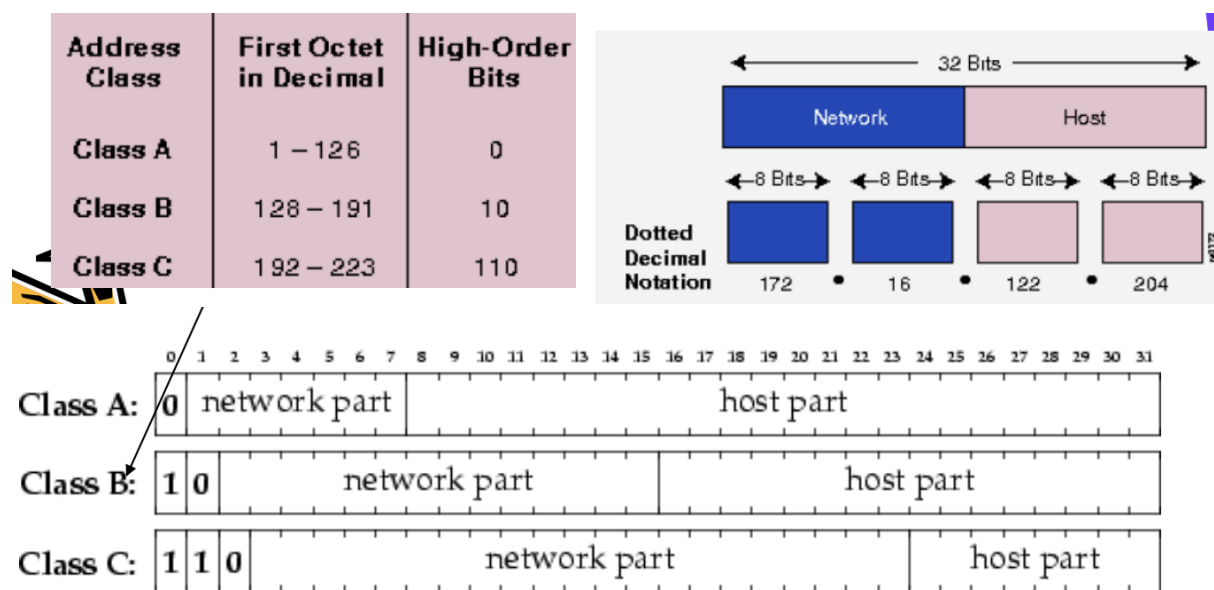
- packetization: information sent or transmitted across the internet which is reformatted and broken down into smaller parts (packets)

Internet Protocol (IP) (Addressing)

- unique identifier
- four dotted decimal notation

Identification of a computer:

- uses an IP address (Internet protocol address)
- system uses a domain name server (DNS) that converts text-based addresses to IP addresses



Ipconfig: allows you to get the IP address of a Windows computer (ipconfig /all)

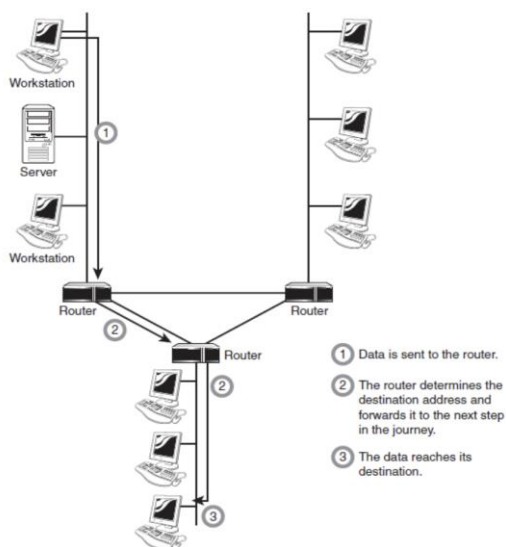
Ping: utility software to measure responsiveness, how quickly data can reach a server and back

Network Latency

- thing that Ping measures
- elapsed time for data to make a round trip from point A to point B and back. milliseconds.
- 40~200ms is all acceptable

Routers

- a computer device that forwards data packets toward their destinations
- junction between 2 or more networks to transfer
- redirect traffic to another route



Internet Service Provider (ISP)

- provide a path or connection for individuals to access the internet such as China Telecom, 电信公司 basically

Connection Devices

- Telephones send/receive analog signals
- Computers send/receive digital signals
- you need modem to convert from one to another, modulator-demodulator (modem)

Data speed

- the speed that modems work at is called transfer rate kbps

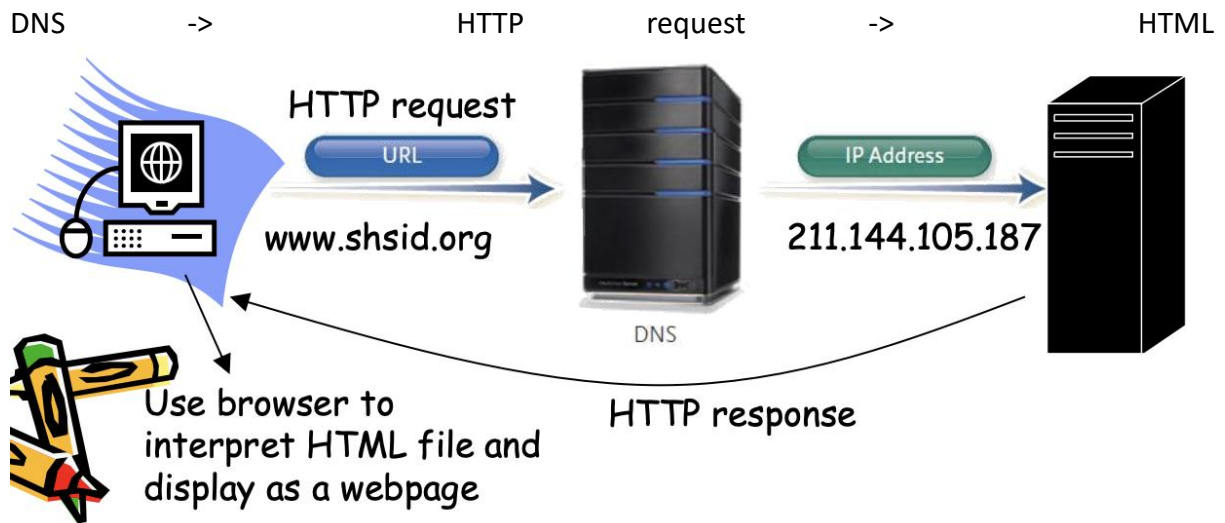
Browser: programs that provide access to web resources (google!)

- browsers connect people to resources with specific location/address

Uniform Resource Locators (URLs)

- two basic parts, protocol (http://)

- domain name (google.com)
- .com is top level domain (along with .edu, .net and .gov)



Web 1.0: focused on linking existing information

Web 2.0: support more dynamic content creation and social interaction (like facebook and twitter)

Web 3.0: focuses on computer-generated information

Email and Spam

- email: electronic message sent over the internet
- spam: unwelcome email
- SMTP Simple Mail Transfer Protocol used for sending email messages from one email account to another on the internet (sending)
- IMPA internet message access protocol allows email clients to retrieve email from a remote server (receiving)

Instant messaging

- allows two or more people to contact each other via direct, live communication

Search engines: specialised programs to help you find information/resources on the web/internet

- special programs called spiders continually look for new information and update the databases

web directories/subject directories

- organised information according to categories or topics

metasearch engines:

- automatically submit your search request to several search engines simultaneously

electronic commerce types

- B2C is business to consumer, online banking, online stock trading, online shopping etc.
- C2C is consumer to consumer such as 闲鱼, depop etc.
- B2B is business to business, involves one business to another like sourcing or supplying

cloud computing: uses the internet and web to shift activities of multiple computers onto another computer

- cloud suites are available anywhere you can access the internet and stored at a server in the internet
- cloud printing enables printers to be accessed through cloud computer
- cloud storage, virtual storage

web utilities:

- plug-ins are programs that automatically started and operate as part of your browser
- filters block access to selected sites which monitor use and generate reports
- file transfer utility: copy files from computer to special config servers (downloading)
- copy files from your computer to another computer (upload)
- uses file transfer protocol (FTP) and secure transfer protocol (SFTP)

Files

`f = open(file name, file mode)`

`r` -> reads content

`w` -> overwrites text in pre-existing file

`a` -> append the text to the end of the existing file

`read()` -> reads entire file content and returns it as a string

`f.close()` -> closes file

`f.write(string)`

`f.read(string)`

Privacy, Security and Ethics

privacy issues

- accuracy -> is the data correct
- property -> who owns the data
- access -> responsibility of those who control/use data

large databases

- industry of information resellers/information brokers
- personal information could become marketable commodity
- issue:
 - collecting public, but personally identifying information
 - spreading information without consent
 - spreading misinformation

Ip addresses: can trace the internet activities to their origin from the history file

cookies -> small pieces of information that are deposited on your hard disk from websites you visit

traditional cookies provide information to a single site, ad network/adware cookies record your activities from multiple sites

spyware: a wide range of programs that are designed to secretly record and report an individual's activities on the internet

web bugs are small images /html code hidden within an email message

computer monitoring software -> most invasive and dangerous type of software

category of programs known as antispyware or spy removal programs are designed to detect and remove cookies, web bugs and monitoring software

security: personal security and computer security'

computer crimes

- malicious programs
 - viruses: programs that migrate through networks and operating systems and attach themselves to different programs and databases
 - worms: do not attach themselves, it's self-replicating, clogging the system so that it's stopped and slowed
 - trojan horses: programs that come into a computer disguised as something else such as freeware, computer games, free antivirus programs
 - zombies are computers that are infected, a collection of zombie computers is called a botnet or robot network
- denial of service
 - DoS attack attempts to slow down or stop a computer system by flooding a computer with requests for info and data
- internet scams
 - it's a scam on the internet, phishing is to trick users into thinking a fake but official looking website is very legitimate

measures to protect computer security

- restricting access -> passwords/biometric scanning

- encrypting data
- anticipating disasters (disaster recovery plan)
- preventing data loss (RAID)

types of computer criminals

- hackers create/improve programs to share with fellow hackers (typically not criminals)
- crackers create and share programs designed to gain unauthorized access to computer systems or disrupt networks

ethics are a standards of moral conduct

copyright is the legal concept that content creators get the control use and distribution of work

software piracy is unauthorized copying and distributing of software

plagiarism is representing some other person's work and ideas as your own without giving credit

2D List

A 2D list (two-dimensional list) is a list of lists.

It can represent structures like tables, grids, or matrices.

Accessing Elements

1 2 3

4 5 6

7 8 9

Use double indices: `nums[row][column]`

Indexing starts at 0.

Negative indices (-1) access elements from the end.

`print(nums[2][1])` # Accesses 8

`print(nums[-1][-1])` # Last row, last column: 9

Measuring Size

`len(nums)` → number of rows

`len(nums[0])` → number of columns (assuming all rows have equal length)

5. Initializing a 2D List Using Loops

```
nums = []  
for i in range(x):  
    row = []  
    for j in range(x):  
        row.append(i)  
    nums.append(row)
```

Simplified Initialization

```
nums = [[0 for i in range(4)] for j in range(4)]
```

Printing 2D Lists

Row by row:

```
for row in nums:  
    print(' '.join(str(val).zfill(2) for val in row))
```

Column by column

```
for i in range(len(nums[0])):  
    for j in range(len(nums)):  
        print(str(nums[j][i]).zfill(2), end=' ')  
    print()
```

Databases and DBMS

two perspectives to view data:

- physical: actual format and location
- logical: meaning, content and context

batch processing

- data is collected over several hours, days and weeks

real-time processing (online processing)

- data is processed at the same time the transaction occurs

data organisation

- character is the most basic logical data element

- field is the next higher level, group of related character
- record is a collection of related fields
- table is a collection of related records
- database is an integrated collection of logically related tables

database:

- data redundancy
- lack of integrity might occur

need for database:

- sharing
- security: users are given passwords or access only to the kind of information they need
- less data redundancy: without a common database, there might be data redundancy
- data integrity: there could be various variations of one source

database management system (DBMS)

- DBMS engine: bridge between logical view and physical view of data
- data definition subsystem: logical structure is defined by data dictionary or schema
- data manipulation subsystem: tools for maintaining and analyzing data
- application generation subsystem: tools to create data entry forms, working with programming language
- data administration subsystem: overall management

development: manual management -> file system -> DBMS

hierarchical database

- nodes are points that connect to places, parent node and several child nodes, one-to-many relationship

network database: a hierarchical arrangement of nodes, many-to-many relationship

relational database: no access down a hierarchy, different tables,

table and its data -> relation

all related tables have a common data item or key field

ER diagram (entity relationship diagram)



types of databases

- individual database/microcomputer database
- company database
- distributed database
- commercial database

database is productivity (yay!) but it needs maintaining which is yay!

Query Languages

- Enable non-programmers to use certain easily understood commands to search and generate reports from a database (basically a search bar)
- Most widely used -> SQL (Structured Query Language)
- It's like simplified coding -> like log in/log out/delete info etc.

Data Hierarchy -> by fields and records

Record -> file (table) -> database

Num, string, boolean, date, string -> field type (primary key that identifies a record)

Record		
Record		
Record/Field	field	field

Four statements:

- Select
- Insert
- Update
- Delete

Select * from **Student** where **Name** = "Hans"

*** = all information, you can also switch this out for specific field names**

Selects all information from the table 'Student' whose name is 'Hans'

Insert into **Student** values (1234, "Jason, "M", CHINA)

The values -> update values of fields

Student -> tables

This inserts all of the following values into the table named 'Students.'

Update Student set **Nationality** = "China" where **Gender** = 'F'

This is the new update.

Gender = 'F' is the condition.

This updates all students whose gender is female as having the nationality of Chinese

Delete from Student where Name = 'Hans'

This is also the condition

This deletes the information of Hans from the table 'Student'

A table can have only one primary key.

Sorting Algorithm

- improves efficiency
- makes data more readable
- (it's like on Amazon where you can sort by prices low to high)
- List all of the elements in a list/array from largest to smallest (or vice versa); It's just an algorithm that puts it in a certain order

types of sorting

- bubble sort: starting from the beginning of the list compare every adjacent pair and swap if they're not in the right position; round #s = n-1 (n=number of values)
 - it kinda looks like:
 - round 1:
 -
 - 95473 -> false, 9>5, change positions
 - 59473
 - 54973
 - 54793
 - 54739
 - round 2:
 - 54739 -> false, 5>4, change positions
 - 45739 -> true, remain the same
 - 45739 -> false, 7>3, change positions
 - 45379 -> true, remain the same
 - etc etc.

Selection Sort:

- Find the smallest unsorted element.
- Swap it with the first unsorted element.
- Repeat for each remaining position.

Steps:

- Find the smallest number in the whole array.
- Swap it with the first element.
- Find the smallest from remaining list, swap with second element.
- Repeat until fully sorted.

Comparison with Bubble Sort:

- Bubble Sort: Swaps many times each round.
- Selection Sort: Swaps only once per round, after finding the min.

Insertion Sort

- Maintains a sorted sublist on the left.
- Takes one element from the unsorted portion and inserts it into the correct position of the sorted sublist.

Steps:

- Start with the second item.
- Compare it with items before it.
- Shift larger items to the right.
- Insert the item into its correct place.
- Repeat for all items.

Example:

- List: 49 38 65 97 76 13 27 49
- Sort steps:
- Insert 38 into [49] → [38 49]
- Insert 65 → [38 49 65]
- Insert 97 → [38 49 65 97]
- Insert 76 → [38 49 65 76 97]